

1. Objective

Simulate the decrease in salt concentration over time in a single-compartment electrodialysis (ED) batch cell. Two ODE models are compared: a linear model with constant removal rate, and a more realistic exponential model in which removal rate decreases as ion concentration falls.

2. Governing Equations

Both models solve the following ODE at each time step:

Linear (constant rate): $dC/dt = -(I \times \epsilon) / (F \times V)$

Exponential (concentration-dependent): $dC/dt = -(I \times \epsilon \times (C / C_0)) / (F \times V)$

where C is the concentration at time t , C_0 is the initial concentration, ϵ is current efficiency, F is the Faraday constant (96485 C/mol), and V is the volume. Solved using `scipy.integrate.solve_ivp` (Python).

3. Parameters

Parameter	Symbol	Value
Applied current	I	0.5 A
Current efficiency	ϵ	0.9 (90%)
Volume of water	V	1 L (0.001 m ³)
Initial concentration	C_0	50 mol/m ³
Simulation time	t	60 min (3600 s)
Faraday constant	F	96485 C/mol

4. Results – Concentration vs. Time

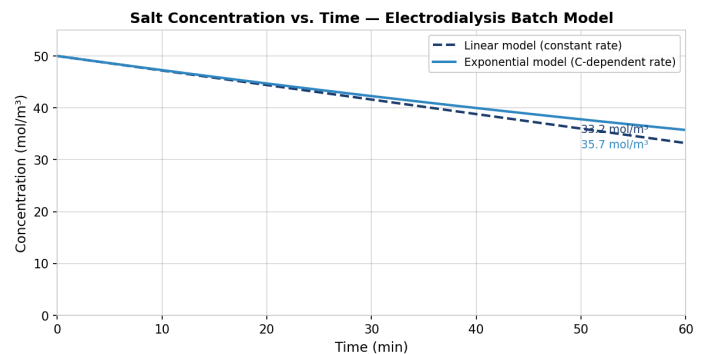


Figure 1: Concentration vs. time. Linear model (navy dashed), exponential model (blue solid).

$I = 0.5$ A, $\epsilon = 0.9$, $V = 1$ L, $C_0 = 50$ mol/m³, $t = 60$ min.

5. Quantitative Summary

Model	Final Conc. (mol/m ³)	Salt Removed (%)
Linear model (constant rate)	33.2	33.6%
Exponential model (C-dependent)	35.7	28.5%

6. Discussion

The linear model predicts a straight-line decline from 50 mol/m³ to 33.2 mol/m³ (33.6% removal) after 60 minutes at 0.5 A. The exponential model is more accurate: as concentration drops, fewer ions reach the membrane, weakening the driving force (28.5% removal). This framework can be extended to multi-ion transport, membrane selectivity, and concentration polarisation. This work directly supports the PhD project's goal of advanced model development for electrodialysis and bipolar membrane processes for salt and nutrient removal.

7. Implementation

Code written in Python using NumPy, SciPy (`scipy.integrate.solve_ivp`), and Matplotlib. An interactive simulator was also built in React (recharts) for real-time parameter exploration. Full source code available upon request.